



Web-Based Rental Building Management System

Software Design Description Document

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Abstract

Rental Building Management System is a web based project through which a user can easily access all the rental houses around his workplace in just a few clicks. Customarily, people have to contact the brokers for finding the rental homes and also have to visit several places. This is a tedious process and can take a lot of time but our web page will create a direct link between a user and the owner with transparency.

As we all can see from the stats shown above, there are still a number of people who do not have their own homes and are accommodating in rental building. Many people have to migrate from their residential area to some other place for a particular slot of time so they prefer to rent houses, PGs etc. People face difficulty finding places to stay in a city they are unaware of, so people will use our Rental Building management platform for finding such assured places at their convenience.

Many people of different professions like students, working executive, households etc. faces a change of residence and have to shift to that particular area where they are needed, but finding a good and healthy house or property is a big task for them our Rental Building management system will play a major role in these type of things and will help such citizens and made their life easier and simple.

Keywords: HTML; CSS; Java Script; UI; Web Application

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Chapter 1

1.1 Project objective

- i. To develop a Rental Rental Building Management System that allows users to view customers as well as data recorded of houses and hostels.
- ii. It is a system that enables editing or saving the user and customers' data where they can edit and see their records also.
- iii. By the increasing number of parental guidance or hostels.
- iv. There are some people that are not even able to afford it and some for personal issues.
- v. So, to produce and secure sufficient supply of affordable rental Rental Building to meet further needs. To provide a variety of house types, cost and location in cities and ensure too took a living for you.
- vi. To development of Rental Building Management Website that is easy to access and secure.

1.2 Scope of the project

If we talk about the scope of the Rental Building management system then the biggest reason for its impact is the rapidly growing cities and societies which will create bigger opportunities for the Rental Building management. Residents or clients need good and affordable rental according to their needs, in which Rental Building management will play a major role. The system will sustain because of the user, their demand and their contributions. We will upgrade the system time to time, will add more features as per the user convenience.

User has to select his category and fill in a few details along with his mobile number and email ID. Then the OTP will be generated and the user has to enter that OTP and create a password to register. For log-in, the user has to use his mobile number or email-ID along with the password.

After Log-In users will have complete details along with the images of all the available rental flats and houses around the location which he has chosen. We will also provide the categories which will further

help the user to find the place to live as per his convenience.

Once the user finds the place which fulfills his needs, he can directly contact the owner. There will be one more available option for the owner who can also add new rental homes as per availability. So our webpage will act as a broker between the owner and the tenant with no extra cost and transparency. Both the parties will be in direct contact and this will help in saving time and money for the users. Our webpage will use technologies such as HTML, CSS, and JavaScript. We will add new and powerful features to our web page which will provide easy user friendly access and an attractive interface.

1.3 Development

As the user opens the web page there will be two options to register and login. When the user clicks the register option a new tab will open with two options. Register as tenant or register as owner.

or users can easily access without any advertisement while using our website also shows you the contact details of the rental apartment owner which make it easier to comfortably talk to the rental apartment owner.

- I. Help support Chatbot
- II. 24x7 accessible to the website
- III. Show result on the basis of user preference.

1.4 Stakeholders

Our application can be used by anyone such as elders, students, etc. Because it does not have any limitation except the authentication of the user that he/she is a minimum of 18 years old. It requires less memory to install and is easy to operate, which is immensely helpful to the user. It only requires your sign-up when you want to see the seller's details or the house owner's details to contact. It is extremely helpful for those people who are unaware of the new place where he/she has not lived in here. It also provides you with a GPS location page which helps you to navigate to your destination safely.

1.5 Role of authors

Dr. Rohit Rastogi played his role as project guide and mentored us on how to find the problem statement and contributed in project making with ensuring that the quality of the content written is profound. Mr. Prabhinav took the task of analyzing the data and contributed to the technical part. Mr. Rayush did the interpretation of data and Mr. Prabhinav and Mr. Rayush did the conception and design of the project. Mr. Prateek handled the task of assembling the literature survey with all the graphical presentations.

1.6 Gantt chart

The above given Gantt Chart shows how the team managed all the work that has to be done in the given time limit of 16 weeks. The first few weeks were spent in the identification of the problem statement and then we worked on the technical and coding part of the project with report writing in continuation. In the last few weeks we analyzed the result and tested whether our application works effectively or not (as per Figure 1).

Chapter 2

2.1 What is rental Rental Building management system

Rental Building management comprises two separate words one is Rental Building and the other is management, as we can see in simpler terms managing houses is what should be called as Rental Building management but it has a broader perspective. Traditionally, Rental Building management was all about rent collecting, allocation and maintenance but if we talk about modern day scenarios many small tasks are also added in it like interactive sites, community involvement etc. So to tackle all these things we came up with an online Hous-

ing Management System in which the user will be free to explore various properties, rental houses etc. at their convenience which is an astonishing thing. The client can book their favorite property online with just a few clicks and they can also directly talk to the owner with details provided at their end

The above image shows us beautifully how we have taken a leap from normal Rental Building management to an online Rental Building management system. The house is connected to the laptop with a cable and the image of the house is shown in the laptop screen which shows that the house is up for rent or sale via online Rental Building management system (as per Figure 2)



Figure 1: Gantt Chart showing the project plan of our team.

2.2 Global records and statistics on RHMS

In the earlier times all the records were maintained in registers and booklets which was a very hideous task.

But after the introduction of the online system the work has become easier and better. Earlier, people must be present physically at places if they want to ensure the status and quality of the property they are going to rent or buy but after the introduction of the online system all these things can be checked through online mode. Due to this much convenience many countries are using this online Rental Building management system. If we see the records then an approximate of 30% people live on rent in India, 36% in the United States, 11% in Russia and many more countries. All these countries use online Rental Building or property management systems for making their work easier (Voumick, D., *et al.* 2021) [7].



Figure 3: Countries with most renters.

The above image shows us the statistics of how much renters are there in many different countries from the lowest to highest. The global analysis has shown that the U.S. and U.K. has shown a major growth in the number of households renting their homes which in turn results in the downturn of the country's economy (as per Figure 3).

2.3 Automation and its need of HRMS in 21st century smart society

Rental Rental Building Management System Automation is needed in the 21st century for saving time, reducing human errors, and doing your work efficiently. Nowadays automation is in our daily lifestyle which we also see in our daily work like using applications and phones. The need for Rental Rental Building Management System to save or record data carefully and It's a type of data collection where there is no chance of duplication where every data is stored on an online server where users are able to see it at any time. A Rental Rental Building Management System is also needed because of the daily increase of hostels that take charge of insignificant students which also resolved their problem and in our RMS you can easily open it with one web browser... It also shows you the nearby location according to your need and provides you various services like GPS,

Retail food delivery, etc. In the 21st Century, it saves a lot of money and time (Gommans, H.P., *et al.* 2014) [2].



2.4 Need of Rental Building management in 21st century Society

If we look at the present scenario, due to the decrease in the availability of land, there is a hike in the price of properties as well as rental homes especially in smart cities like Mumbai, Delhi, Bangalore and others. People from all over the country visit these places for jobs or for educational purposes and the wages of the people is less as compared to rent and also due to the less availability of the land the people also don't know about the unknown places and places with less rent (Kirmani, J.A., *et al.* 2017) [\[4\]](#).



Figure 4: Finding the perfect house for yourself.

Source URL: https://www.baymgmtgroup.com/wp-content/uploads/2021/04/shutterstock_92285236-min-400x400.jpg

The above image demonstrates that with the help of an auto- mated rental house management system one can make a checklist of his/her demands and will sort out the best match from the online application which will save their time and money (as per Figure 5).

websites in these sectors such as 99 acres, Magicbricks are not well managed, So they came up with a new system which can simplify the work for its users and all their work such as rental house search and property dealings can be effective.

The system is operated on the Django framework with a user- friendly interface. When a user opens the application, the user must sign-up and log-in. If he/she wants to talk to the house owner and if someone

wants to post their property online then they have to log-in as an owner on the application. The application is developed using the following:

- I. Django
- II. HTML CSS
- III. Java Scripting
- IV. Python
- V. MySQL

The outcome of the project was that they developed an application that has improved the lifestyle of the people as they can easily find rental property. The owners on the other side can find tenants for their rental property. This application works as a matchmaker for both the tenants and the owners. The Web based Rental Housing Management System is an advantage for both tenants and landlords (Rathore, K., *et al.* 2021) [6].

Chapter 3

3.1 The Impact of Rental Building management system in country's development

Our webpage will provide a user friendly system that will be useful to all the classes of the society. So, it will help provide shelter to all the category of users. Hence it will be helpful to improve the quality of life of people and will also lead to the economic and social development of the nation. Our online rental Rental Building management system will be very helpful for the citizens of the country in accommodating the house and property they want with proper chat and concern of the owner which will lead to minimum disputes between the citizens (Kirmani, J.A., *et al.* 2017) [\[4\]](#).



Figure 6: Satisfied user shaking hands with the property owner. Source URL:
<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSD8efgtUFD2KEHirSVESH5ooZy30hn3yN264w8usqp=CAU>

The outcome of the project was that they developed an application that has improved the lifestyle of the people as they can easily find rental property. The owners on the other side can find tenants for their rental property. This application works as a matchmaker for both the tenants and the owners. The Web based Rental Housing Management System is an advantage for both tenants and land-lords (Rathore, K., *et al.* 2021)

M.R., *et al.* (2021) and his team demonstrated that a house is a place where people gather, they can improve their lifestyle, and can grow economically as well as socially. Their team depicted that initially buying a house is a big task for an individual because it needs a very high capital amount.

3.2 Literature Review

Rathore, K., et al. (2021) and his team demonstrated a Rental House Management System. They designed an online house renting system for the ease of the citizens. They found that nowadays finding an affordable renting house is a major issue and the leading

were Hypertext markup language (HTML), CSS, JS, Hypertext Pre-processor (PHP) and MySQL database.

This is a web based software which is used for making the renting easier for the user. The team has used the cloud service with components such as app server, payment engine, wallet engine, chat server, message engine and authentication server. Centralized databases using MYSQL and programming languages are angular js, PHP and laravel. A cloud based mobile application with user friendly interface and secure environment is established.

This is a database type system that is able to access all the information and is to manage and it is especially built for rental home owners.

- I. HTML
- II. CSS
- III. JavaScript
- IV. PHP

This project is a database type software that is used to help landlords to manage the records or pay history of tenants that's resolved the problem of storing records and also calculate the profits (Gommans, H.P., *et al.* 2014) [2].

3.3 Setup Steps

- i. Initially we started with the identification of the problem statement and analyzing the major faults and issues.
- ii. Our team asked the opinion of various people on the current management system and asked if there was any kind of issue.
- iii. We collected data from the current working websites on renting management systems and seek for any type of issue caused.
- iv. We talked to various owners and their requirements like what they need for a better experience on the website.
- v. After collecting all the required data all the team members analyzed it, and started its interpretation.
- vi. Then we start to learn HTML and CSS for making the front end of our web based application.
- vii. After that we learned the languages required for making the back-end of our rental house management software.

After gaining all the knowledge required the team divided the work and started to make the web application

3.4 Use case diagram



Figure 7: Basic use case diagram showing how the Rental House Management System works.

Chapter 4

4.1 Methodology and setup of experiment

For the proper working of our web application we collected data from many resources. We talked to people (customers) so that we can understand the problems they are facing in the renting system. We also collected views of people on the current online websites which provide an online renting system and noted down the problems faced by them or anything that caused any kind of inconvenience, so that we can enhance the quality of our work.

The methodology that they used to create in their Online Rental Management System is Structured Query Language (SQL), PHP will be used to collect or store the data in a database. And for web design they used html, css, bootstraps. It consists of three parts: Client, Web Server, Database Server.

Afzal, S., *et al.* (2021) and his team demonstrated that if you are a stranger in some city and wanted to rent a house. Then, it is very difficult to find a suitable house at a time that meets your requirements. The main motive to build an Online Rental Management System. So, this website helps users to register individual homes or apartments to help you in finding the perfect rental home that is also used for property selling, leasing, and buying. Which makes it easier for tenants to choose a desired house or property within budget and Landlords can easily sell their property by uploading and registering the website.

Once the user finds the place which fulfills his needs, he can directly contact the owner. There will be one more available option for the owner who can also add new rental homes as per availability. So our

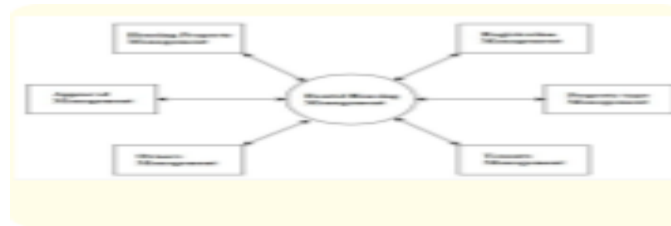
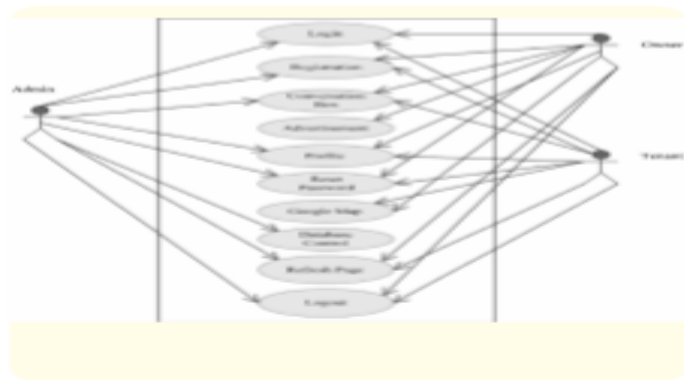
webpage will act as a broker between the owner and the tenant with no extra cost and transparency. Both the parties will be in direct contact and this will help in saving time and money for the users.

4.2 Rental Building Management System

Paper Title and Author	Introduction	Methodology	Data Set and Algorithm	Conclusion	Future Scope
Rental House Management System (Rathore, K., <i>et al.</i> 2021) [3].	team built a web based application which will make the online renting system more efficient and convenient. They resolved the issue of difficulty in finding affordable rental property.	The user opens the web application he must log-in and sign-up if he wants to talk to the owner and the same goes for the owner side if he wants to put his property on the web application.	MySQL Database on phpMyAdmin. Cache server on Xampp. P.NET using c#.	The outcome of the project was that they developed a web application that made the user lifestyle easier in finding rental houses and the owners can also find the tenants of their liking for the rental property.	rent scenario there are a number of people who are living in rental houses and are finding rentals to live in so this web application will make their task of finding rental property easier and more convenient.
House Rental Management System (Misyam, M.R., <i>et al.</i> 2021) [5].	Their team built up a web based application which will help the customers to find the nearby rental properties and they also made sure that there is a proper gateway of conversation between the parties.	Their software works on a very simple method, the user has to login and have to find the proper rental house which meets their demands and can contact the owner of the property for further details.	The languages used to build their application were HTML, CSS, JS, PHP database used was MySQL database.	They concluded that they have finally built a software that meets all the requirements according to the demands and it will surely help in enhancing the lifestyle of the people.	house is not an easy task for an individual hence one has to first rent a house and then save up for buying a house. So, their web app will help such people who are in need of a convenient rental house. They will also add up new features for enhancing the user experience.

al Rental Building Management irmani, J.A., <i>et al.</i> 2017) [4].	This is a software ASP.NET. The base idea of this research is to provide “affordable” shelter to all the classes whether lower upper or middle class.	ware has user specified searching and direct contact between buyer and seller. Cart system is also provided. Easy addition, deletion and searching of houses with user -friendly interface.	ASP.NET Data- base used is SQP 2008. IS and the .NET framework	ine software provides exact information and can be accessed easily from any- where. cart system and mail based information increases its demand in the market.	This software uses 2010 .NET version which is now upgraded to .NET 2015 version. own searching and satel- lite/map based searching can also be added in this software.
3ased Rent- al Rental Building Man- agement System (Ikuomola,A.J., <i>et al.</i> 2020) [3].	This is a web based software which is used for making the renting easier for the user.	The team has used the cloud service with compo- nents such as app server, payment engine, wallet engine, chat server, mes- sage engine and authenti- cation server.	Centralized databases using MYSQL and programming languages are angular J.s., PHP and Laravel.	A cloud based mobile applica- tion with user friendly inter- face and secure environment is established.	ed system can be used i.e. if the tenant does not have a device to contact the owner then he/she can use the smart door technique. House rent countdown system can also be added to remind the tenant of the rent.
al Rental Building Management n (Gommans,H.P., <i>et al.</i> 2014) [2].	Most of the work done by the Rental Building managers is on paper and maintaining records is re- ally difficult. So their webpage will basically help the user to maintain data in an organized way and will also prove a user friendly interface for both the owners and the tenants.	ou have to log in then you have to select an option that is visible on the website. It will show how the data is collected from the user through the system that has been kept in the system database.	system design, frame- work, er address, etc.	This software is used as an inventory to provide a frame- work that enables the manag- ers to make only reasonable transactions made within a limited time of the framework.	This system software has to also give a pay slip and data that are directly connected to the Microsoft Access that automati- cally sends the slip of recorded payment to the tenants.

Online Rental Rental Building Management System (Afzal, S., <i>et al.</i> 2021) [1].	m make a website of Online Rental Rental Building Management System which is built on the basis of eliminate paper based word, save time and give an desirable result according to the user need	Firstly, they used web servers like SQL and PHP that enable the website to store or collect data in a database. And they used HTML, CSS, Bootstraps. That contains three parts: client, web server, database server that help to build an online website.	Structured Language (SQL), PHP, HTML CSS, Query:bsite is used for tenants, landlords, hostel, PG, etc. It is more flexible than the existing system. It has more accuracy and compatibility in terms of saving time and working output. It is an online website where a user has to firstly register and fill out the details then the recommendation of that similar output shows on the web page. It is very beneficial for students.	This is a modern Online Rental Rental Building Management System of Web application that is built by PHP and SQL. And it also has a client/server HTTPS protocol. For the future scope this system has a GPS server which helps to give the location of a house. It is a user-friendly system which makes it easy for a tenant to search a desired house and for landlords which just have to input the details of their properties.
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4.3 Flowchart

The above shown diagram is the detailed flowchart of the Rental House Management System. It shows how the user will enter into our web application where he will sign up and proceed as shown in the above diagram (as per Figure 11).

4.4 Result and Discussion

Here a user can choose whether he/she comes for our website as a tenant or owner. It makes easy to categories between the tenants and the owners. We make it easy so anyone can use it.

4.5 ER Diagram

E.R. diagram

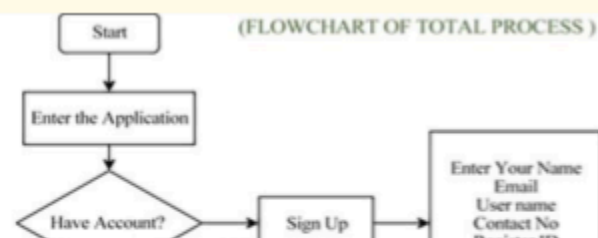


Flow Chart

An ER diagram showing the working of Rental House Management System.

Flow Chart

IV.6



4.6 Future Directions

- To also provide service to a user like personal transportation support which are directly connected to Ola Cabs, Uber where a user can book a cab for its destination which with 2-4% discount.
- To make our online web application also for Android and IOS app systems.
- To add dynamic features and also create an AI Chat bot which helps you to solve your biggest problem regarding transaction, using this online web application.
- To make our application system on voice command also with different languages.

4.7 Conclusions

This Rental Housing Management System is a web application that is made by using HTML, CSS, JS, MySQL. All of this was developed in visual studio code by Microsoft. Our rental house software is a very convenient and user friendly software and it provides its users with various new features for enhancing their experience. The user will enter into our website and first he has to log-in or sign-up with their credentials and then they will be directed to our webpage where they will have to select the region in which they want to rent a house and then after that they will be shown all the available properties in that specific area, from there they can select the house that meet their demands. This rental software will enhance the renter owner relationship and will aim at making customers' lives easier.

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